IBM University Engagement Project Submission

"Exploring the Power of Agentic AI with IBM Granite and LangFlow"

Project Tile – CAPSTONE PROJECT

Domain of Project – ***INDUSTRY***

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OUTLINE

- ***Problem Statement***
- ***Proposed System/Solution***
- ***System Development Approach*** (Technology Used)
- ***Algorithm & Deployment***
- ***Result (Output Image)***
- ***Conclusion***
- ***Github LINK***
- ***Future Scope***
- ***Refrences***

PROBLEM STATEMENT

Problem statement No.39 – Predictive Maintenanceof Industrial Machinery

Example: This project aims to predict equipment failures in industrial machines before they happen using sensor data. By analyzing real-time

parameters like temperature, torque, and tool wear, a classification model is built to identify issues like tool wear, overheating, or power failure. This helps enable proactive maintenance, minimizing downtime and operational costs.

PROPOSED SOLUTION

- Theproposedsystemaimstoaddressthechallengeofpredictingpotentialmachine failures in industrial environments before they occur. This involves leveraging dataanalyticsandmachinelearningtechniquestoaccuratelyforecastfailurepatternsbasedonoperationaldata.The

  solution will consist of the following components:

- Data Collection:
  - Gather historical sensor data from industrial machines, including parameters like temperature, torque, rotational speed, and tool wear.
  - Focus solely on the available sensor readings from the dataset without relying on external contextual inputs.
- Data Preprocessing:
  - Clean and preprocess the raw sensor data to handle missing values, outliers, and inconsistencies.
  - Perform feature engineering to derive relevant indicators (e.g., rate of temperature change, wear thresholds) that may signal upcoming failures.
- Machine Learning Algorithm:
  - IBM watsonx.ai'sAutoAIgenerated a Snap Random Forest Classifier (Batched Tree Ensemble) that achieved a cross-validation accuracy of 99%. This ensemble model aggregates multiple decision trees to classify failure types accurately based on input sensor data.
  - The model pipeline was automatically enhanced using Feature Engineering (FE) and Hyperparameter Optimization (HPO) to improve accuracy androbustness. Real-time features like air temperature, process temperature, torque, and tool wear were key inputs for classification.

- Deployment:
  - IBMWatsonx.aiStudio isusedtodeploythe trained Snap Random Forest Classifier, enabling real-time predictions via an easy-to-use interface that accepts JSON or CSV inputs.
  - IBM's hybrid pipeline, which provides quick, scalable, and dependable access to real-time machine failure predictions, powers the model. It also has a public REST API.

- Evaluation:
  - During AutoAItraining in IBM Watsonx.ai Studio, accuracy metrics and cross-validation were used to evaluate the model's performance.
  - Accuracymetrics and cross-validation were used to evaluate the model's performance during AutoAItraining in IBM Watsonx.ai Studio.
  - Result: Thedeployed model accurately predicts failure types such as tool wear, power failure, and overheating, enabling proactive maintenance

SYSTEM APPROACH

Thissectionoutlinesthe overall strategyandmethodology for developing and implementing the Predictive Maintenance System using IBM watsonx.ai Studio.Here's a suggested structure for this section:

System requirements Tobuildanddeploythepredictivemaintenance classification model, the following components were required:

- IBM watsonx.ai Studio for AutoAIpipeline generation and deployment.
- IBM watsonx.ai Runtime for inferencing and model training.
- Watson Machine Learning services for deployment and scalability.
- Online deployment environment with a REST API endpoint for real-time prediction.
- 10–20 Capacity Unit-Hours (CUH)
  Compute environments (1–4 vCPUs, 4–16 GB RAM)

- **Library required to build the model:** Key libraries and tools used in the AutoAIpipeline and backend processing:

  - AutoAI–Automated data preprocessing, feature engineering, and model selection
  - Snap Random Forest Classifier –Chosen as the top-performing algorithm
  - Pandas & NumPy –For initial dataset formatting
  - Sklearnmetrics(internallyused)–Formodelevaluation (accuracy, precision, recall, F1)
  - DeploymentSpaceonwatsonx.ai–Forreal-timeonline predictions via JSON input
  - APIreference&testinginterface–ProvidedbyIBM watsonx.ai for integration

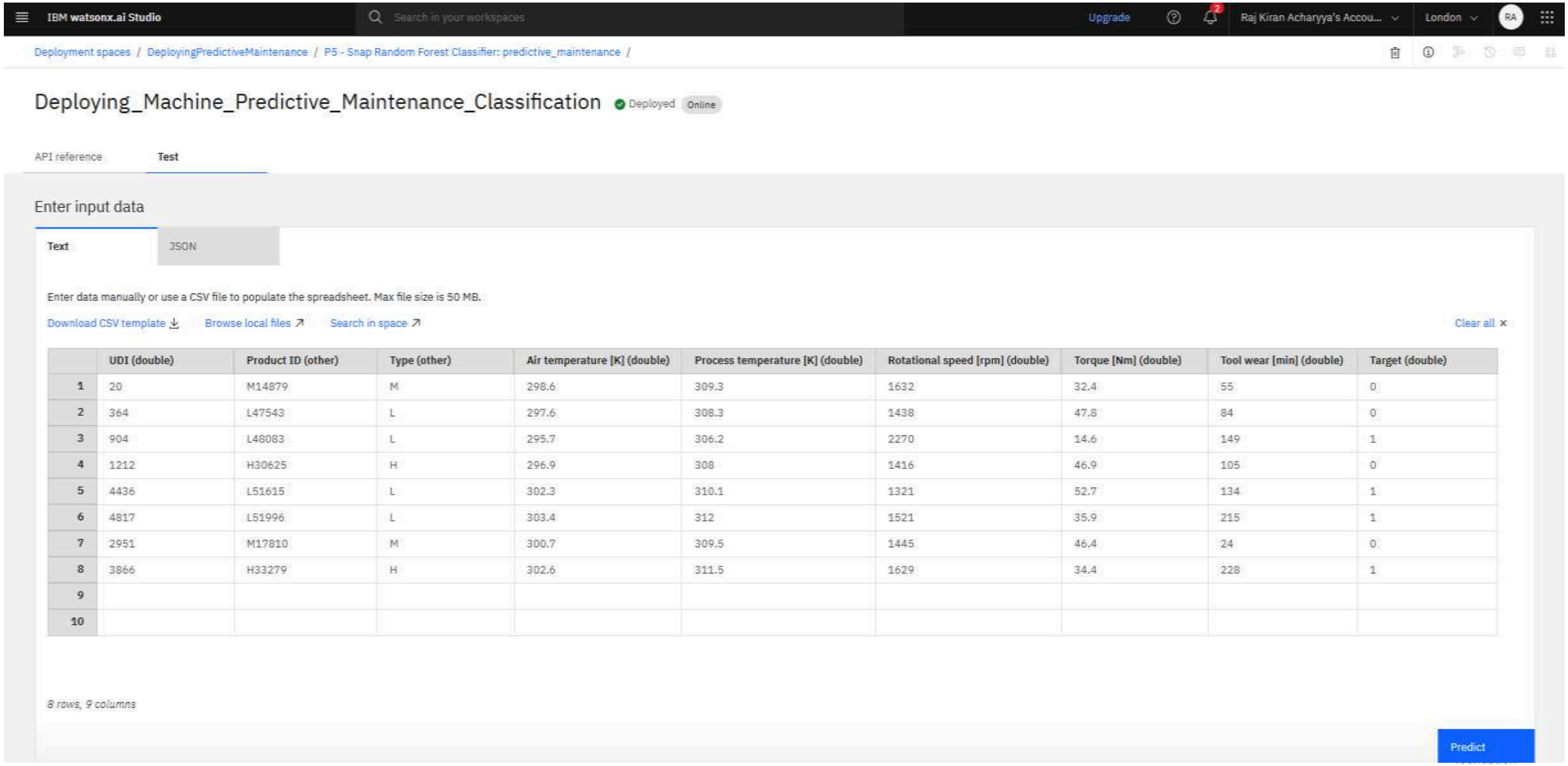
ALGORITHM & DEPLOYMENT

- The SnapRandomForest Classifier,chosen byIBM watsonx.ai'sAutoAIfor itshigh accuracy, it was deployed to deliver real-time failure predictions in the maintenance system. Its robust performance made it ideal for production use.
- Algorithm Selection:
  - ThemodelchosenbyIBMwatsonx.ai’sAutoAIwastheSnapRandomForestClassifier,whichemergedasthebest-performing algorithm during experimentation.
  - This ensemble-based tree classifier is well-suited for handling tabular sensor data and performs well in classifying different failure types. Its ability to manage non-linear relationships and categorical variables makes it ideal for predictive maintenance tasks.
- Data Input:
  - The model uses key sensor inputs—machine type, air and process temperatures, rotational speed, torque, and tool wear—to assess equipment health and predict failures.
  -

These features reflect the machine’s operational state and help the model detect early signs of issues before they escalate.

- Training Process:
  - Training was done via IBM watsonx.ai AutoAI, which automatically handled feature engineering, cross-validation, and hyperparameter tuning.
  - It generated multiple pipelines, with the best one (Snap Random Forest) achieving ~99.5% accuracy.
- Prediction Process:
  - It accepts input as a structured JSON or CSV format through the online deployment interface in watsonx.ai.
  - The model then returns the predicted failure type, enabling preventive action before actual breakdowns occur.
  - Once deployed, the trained Snap Random Forest model classifies real-time operational data into potential failure categories.

RESULT



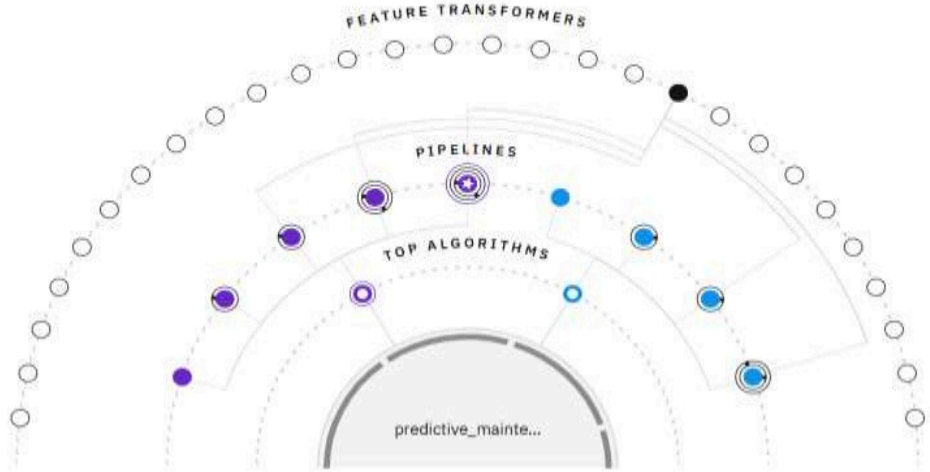
	UDI (double)	Product ID (other)	Type (other)	Air temperature [K] (double)	Process temperature [K] (double)	Rotational speed [rpm] (double)	Torque [Nm] (double)	Tool wear [min] (double)	Target (double)
1	20	M14879	M	298.6	309.3	1632	32.4	55	0
2	364	L47543	L	297.6	308.3	1438	47.8	84	0
3	904	L48083	L	295.7	306.2	2270	14.6	149	1
4	1212	H30625	H	296.9	308	1416	46.9	105	0
5	4436	L51615	L	302.3	310.1	1321	52.7	134	1
6	4817	L51996	L	303.4	312	1521	35.9	215	1
7	2951	M17810	M	300.7	309.5	1445	46.4	24	0
8	3866	H33279	H	302.6	311.5	1629	34.4	228	1
9									
10									

Experiment summary | Pipeline comparison

★ Rank by: Accuracy (Optimized) | Cross validation score

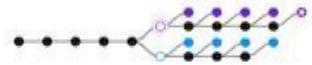
Relationship map

Prediction column: Failure Type



Progress map

Swap view



Experiment completed

9 PIPELINES GENERATED

9 pipelines generated from algorithms. See pipeline leaderboard below for more detail.

Time elapsed: 3 minutes

View log | Save code

Pipeline leaderboard

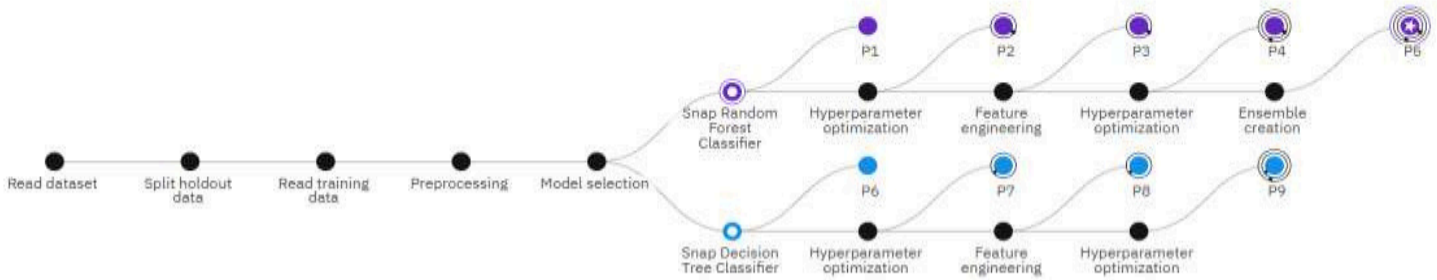
	Rank	Name	Algorithm	Specialization	Accuracy (Optimized) Cross Validation	Enhancements	Build time
★	1	Pipeline 5	Batched Tree Ensemble Classifier (Snap Random Forest Classifier)	INCR	0.995	HPO-1 FE HPO-2 BATCH	00:00:47
	2	Pipeline 4	Snap Random Forest Classifier		0.995	HPO-1 FE HPO-2	00:00:44
	3	Pipeline 3	Snap Random Forest Classifier		0.995	HPO-1 FE	00:00:35
	4	Pipeline 9	Snap Decision Tree Classifier		0.994	HPO-1 FE HPO-2	00:00:04
	5	Pipeline 2	Snap Random Forest Classifier		0.994	HPO-1	00:00:09
	6	Pipeline 1	Snap Random Forest Classifier		0.994	*None*	00:00:05

Experiment summary | Pipeline comparison

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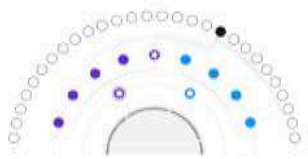
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Swap view



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Prediction results

Prediction type

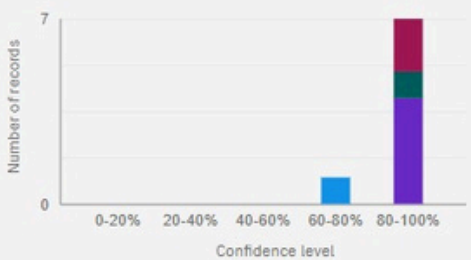
Multiclass classification

Prediction percentage



No Failure
Power Failure
Heat Dissipation Failure
Tool Wear Failure

Confidence level distribution



No Failure
Power Failure
Heat Dissipation Failure
Tool Wear Failure

Display format for prediction results

(•) Table view ( ) JSON view

Show input data

	Prediction	Confidence
1	No Failure	100%
2	No Failure	100%
3	Power Failure	70%
4	No Failure	100%
5	Heat Dissipation Failure	100%
6	Tool Wear Failure	90%
7	No Failure	100%
8	Tool Wear Failure	100%
9		
10		
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14		
15		
16		

CONCLUSION

- In this project, I developed a predictive maintenance model for industrial machinery using IBM watsonx.ai Studio and AutoAI. By working with real-time sensor data and applying the Snap Random Forest Classifier, I was able to accurately classify failure types such as tool wear, overheating, and power failure. The model showed strong predictive performance, making it suitable for proactive maintenance that can reduce downtime and operational costs.
- Through thiswork, I've seen firsthand howAIcan playapowerfulroleinimprovingmachine reliability and efficiency. This project helped me understand the value of predictive analytics in industrial settings and how intelligent systemscan supportsmarterdecision-making.

Git Hub Link

- *https://github.com/Be-mayank/AICTE-IBM-INTERNSHIP.git*

FUTURE SCOPE

- Integration with IoT Platforms:Real-time sensor data can be streamed using IoT devices to enhance live monitoring and quicker failure detection.
- Model Enhancement with Deep Learning: Future versions can use LSTM or CNN models for improved accuracy, especially on sequential or time-series data.
- Expansion to Other Industries: The model can be adapted for predictive maintenance in sectors like aviation, automotive, and manufacturing.
- AutomatedAlertSystem:Integrationof the modelwith maintenance scheduling systemscantrigger alerts and maintenance tickets automatically.
- Edge Computing Deployment: For faster, on-site processing without relying on cloud latency, models can be deployed on edge devices.
- Continuous Learning Loop: With more data, the model can be retrained periodically to improve performance and adapt to new failure patterns.
- Incorporating Cost Analysis: Adding a cost-benefit layer to prioritize maintenance actions based on economic impact.

REFERENCES

- ***Kaggle dataset link –https://www.kaggle.com/datasets/shivamb/machine-predictive-maintenance-classification***
- IBM Cloud Documentation--h t t ps://www.ib m.com/cloud/watsonx-ai

IBM CERTIFICATIONS

In recognition of the commitment to achieve professional excellence



MAYANK

- Screenshot/ credlycertificate(getting started with AI)

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence

Issued on: Oct 23, 2026
Issued by: IBM SkillsBuild

Verify: https://www.credly.com/badges/1b8f5e3e-12b2-4c7f-997f-807f1a752027

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IBM CERTIFICATIONS



IBM **SkillsBuild** Completion Certificate

This certificate is presented to

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for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 02 Nov 2026 (GMT) **Learning hours:** 30 mins

Thank You!

Thank you for your time and interest.